

Xuyang Kang

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PROFESSIONAL SUMMARY

Senior Tech Lead with 10+ years of experience in hyperscale backend, distributed systems, and AI infrastructure. Currently focused on Site Reliability Engineering for Google's Agentic AI platform and Gemini Enterprise. Combines deep systems expertise with advanced training in High Performance Computing (HPC) to architect resilient runtime ecosystems, optimize ML workflows, and safeguard mission-critical AI services.

EXPERIENCE

Google London, UK
Senior Software Engineer, SRE 2026 – Present

- Lead site reliability initiatives for Google's Agentic AI platform, ensuring operational readiness and high availability for enterprise AI workloads.
- Architected and executed comprehensive blackbox testing frameworks (load tests, active synthetic probes) covering **XX** Critical User Journeys (CUJs) across Gemini Enterprise and the agentic AI runtime, validating system resiliency to support key launch milestones.
- Safeguarded high availability via 12-hour on-call rotation shifts, triaging real-time alerts and mitigating high-severity production incidents across Agentic AI services.

Google London, UK
Senior Software Engineer, SRE, Tech Lead – Ads Oct 2022 – 2026

- Architected and implemented a customized OLAP engine monitoring **XX** Critical User Journeys (CUJs) across flagship platforms including AdSense, AdMob, and Google Ads Manager for **X** internal customers; mitigated critical visibility gaps in the Ads ecosystem by delivering an aggregated, user-centric monitoring solution that bridges raw system metrics and business-critical reliability journeys.
- Drove the team's technical vision and strategy while identifying and resolving complex reliability issues as a hands-on individual contributor.
- Engineered an LLM-powered multi-agent system to automate internal team onboarding to the customized monitoring infrastructure; reduced onboarding time from weeks to hours and enabled seamless scaling across the entire Google Ads ecosystem.
- Developed AI-driven dashboards and alerting systems to monitor CUJ health; validated AI performance through rigorous backtesting to ensure accuracy.
- Programmatically defined and measured adherence to SRE best practices to ensure systemic reliability.
- Maintained and improved the reliability of Ads Performance Max (PMax) campaigns and the Ads Build pipeline/data infrastructure.
- Safeguarded high availability via 12-hour on-call rotation shifts, mitigating critical incidents with **\$XX,XXX+** revenue and significant user experience impact.

DoorDash Japan
Senior Software Engineer Oct 2021 – Sep 2022

- Led a team of developers delivering key features to drive the growth of DoorDash's APAC business.
- Successfully launched the DashPass annual subscription model in Australia.
- Led the technical integration and launch of the AfterPay & DashPass partnership.
- Improved Search functionality and A/B testing frameworks for the Japanese market.

ExaWizards Japan
Software Engineer, Tech Lead July 2019 – Sep 2021

- Led the development team for the CompanySearch service, taking it from Proof of Concept (PoC) to partner delivery.
- **Search Optimization:** Architected a high-dimensional vector similarity search solution with complex metadata filters; replaced legacy parallel linear scanning with a hybrid query scheduler across PostgreSQL and Elasticsearch, which served as the foundation for my MSc Research.
- Designed and implemented a user management subsystem to scale the service for public use.
- Introduced and scaled Elasticsearch to enable robust text search and filtering capabilities.
- Established a dedicated sub-team to productionize data pipelines and support research initiatives.
- Defined and enforced software engineering best practices within the team.

Indeed Japan
Software Engineer Nov 2017 – July 2019

- Maintained and optimized infrastructure for Indeed's private cloud environment.

Google Beijing → Mountain View (HQ)
Software Engineer Oct 2013 – Oct 2017

- Contributed to high-impact projects including Street View, Android Studio, YouTube Music, and the Go programming language (Golang).

RESEARCH & ACADEMIC PROJECTS

MSc Dissertation: AI-Powered Scaling of ML Workflows University of Edinburgh
Researcher 2025 – 2026

- Developed a framework to bridge the gap between interactive Jupyter Notebook prototyping and production-grade orchestration using **Kubeflow Pipelines**.
- Leveraged **LLMs and Multi-agent patterns** to automate the transformation of monolithic notebooks into modular, scalable pipeline components, reducing manual productionization overhead.
- Evaluated system performance and scalability on the **Edinburgh International Data Facility (EIDF)** HPC infrastructure to support large-scale data science tasks.
- Integrated advanced prompt engineering to ensure code quality, reproducibility, and alignment with MLOps best practices.

MSc Dissertation: Top K vector similarity search with filters University of York
Research Engineer (Project Lead) 2022

- Investigated vector similarity search combined with filters for the exaBase company search dataset.
- Evaluated the query performance of linear scan, faiss, postgresQL and elasticsearch.
- Proposed a custom query scheduler that utilizes postgresql for queries with no or weak filters and elasticsearch for queries with strong filters.
- Measured average query times and analyzed the experiment data utilizing Multi-factor ANOVA.

EDUCATION

University of Edinburgh Graduated 2026

M.Sc. in High Performance Computing with Data Science

Relevant Coursework: Machine Learning at Scale, Accelerated Systems: Principles and Practice (GPU programming with CUDA, OpenMP and SYCL), Parallel Design Patterns, Message-Passing Programming (MPI), Threaded Programming (OpenMP), Performance Programming

University of York Graduated 2022

M.Sc. in Computer Science with Data Analytics

Nankai University Graduated 2013

B.Sc. in Computer Science

CERTIFICATIONS & LEADERSHIP

- **Management Essentials** — Columbia Business School Executive Education 2023
- *Focus:* Developed core competencies in data-driven decision-making, high-impact leadership strategies, and driving cross-functional alignment through influence and social capital.

TECHNICAL SKILLS

- **Programming Languages:** Python, Go (Golang), Java, Scala, C/C++, SQL, Unix Shell Scripting.
- **High Performance Computing:** Distributed-Memory Programming (MPI), Shared-Memory Programming (OpenMP), Accelerated Systems (GPGPU/CUDA), Parallel Design Patterns, Performance Optimization & Profiling.
- **ML & Search Infrastructure:** Kubeflow Pipelines, MLOps, Vector Similarity Search (Faiss), Search Engines (Elasticsearch), Distributed Model Training, LLM Orchestration, Multi-agent Systems.
- **Backend & Infrastructure:** Kubernetes, Docker, Distributed Systems Architecture, Microservices, High-scale OLAP (BigQuery/Dremel), Hyperscale Internal Cloud Infrastructure.
- **Reliability & Observability:** High-scale Monitoring/Telemetry, SLI/SLO/SLA Frameworks, Alerting Systems, Incident Management, Distributed Tracing, Capacity Planning.
- **Full-Stack & Tooling:** React, TypeScript, HTML/CSS, Backend-for-Frontend (BFF) patterns, JWT/Auth management, internal dashboard development.

COMPETENCIES

- **Research Methods:** Experimental Design (A/B Testing, Benchmarking), Quantitative Analysis, Hypothesis Testing, Multi-factor ANOVA.
- **Data Analysis:** Python (Pandas, NumPy, Scikit-learn), Statistical Modeling, Data Visualization, Exploratory Data Analysis (EDA).
- **AI & Technology:** Large Language Models (LLMs), Machine Learning Architecture, Research Workflow Automation.
- **Languages:** English (Professional), Chinese (Native), Japanese (Professional Working Proficiency).